

Ashvin Anilkumar

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Summary

Graduate student specializing in Robotics and AI, passionate about advancing embodied intelligence through robot learning and perception. Experienced in developing autonomous systems that integrate reinforcement learning with real-world robotic control. Skilled in PyTorch, ROS2, and simulation-to-real transfer for manipulation and navigation tasks. Strong foundation in machine learning, with a research focus on scalable perception and learning-based control in complex human environments.

Education

University of Wisconsin-Madison, Master of Science in Electrical and Computer Engineering Aug 2025 - Dec 2026

- *Relevant Coursework:* Computer Vision, Machine Learning, Optimization, Microrobotics

Government Engineering College, Barton Hill, Trivandrum, Bachelor of Technology in Electrical and Electronics Engineering Aug 2019 – July 2023

- GPA: 8.9/10 (First class with distinction)
- *Additional Coursework:* Minor in Information Technology

Work Experience

Engineer, Backend Development(AI/ML), Qburst Technologies - Trivandrum, India Aug 2023 – Aug 2025

AI Search Using RAG & Agentic Flow

Tools Used: FastAPI, Langchain, LlamaIndex, MongoDB, Postgres, AzureDevOps.

- Developed an AI search bar to efficiently retrieve information about the client's influence and contribution across various countries.
- Designed the system for improved response speed and optimized token usage by more than **56%**.
- Increased the accuracy of search results by **90%** and expanded the range of questions it can handle by **85%**.

AI Report Reviewer System

Tools Used: FastAPI, Langchain, AzureOpenAI.

- Built an AI-powered system for automated evaluation and feedback on technical reports of the client.
- Designed a core framework to generate unbiased reviews using **context analysis** and **coherence checks**.
- Developed a custom **scoring algorithm** assessing structure, clarity, and relevance of the document.
- Achieved **89%** similarity with human ratings and an **35%** increase in evaluation accuracy compared to human ratings, all the while reducing the cost of reviewing by **67%**.

Academic Projects

Multiagent Coordination and Decision Making (RoboCup) December 2025 – Present

Tools Used: Isaac Sim, MuJoCo, ROS2, Pytorch

- This project was done as part of the International RoboCup Tournament under Prof. Josiah P. Hanna
- Implement a multiagent, real-time learning protocol using reinforcement learning and robot learning. The work is in progress.

University Rover Challenge: Obstacle Detection and Avoidance Sept 2025 – Present

Tools Used: ROS2, Python, DepthAI

- Developing a real-time perception and mapping algorithm for autonomous navigation on unstructured terrain.
- Implemented depth-based obstacle detection and reactive control for safe traversal using stereo data.

Skills

Programming: Python, C++ , Rust, JavaScript

AI/ML: PyTorch, Scikit-Learn, LangChain

Tools: FastAPI, Docker, Kubernetes, Git, Linux, Azure

Perception: Object Detection, Sensor Fusion, SLAM

Robotics: ROS2, Gazebo, DepthAI, MuJoCo, Isaac Sim

Hardware: Raspberry Pi, STM32, Arduino, ESP32